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—Albert Einstein

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Fig. 11: Water level is below 75%



Fig. 12: Water tank is full

- DC jack to connect the solenoid valve
- 220V, 3-pin connector to connect water pump
- AC 220V, 10A rated 3-pin panel-mount plug and socket for 220V input
- A 10A rated switch

The mechanical assembly is mounted vertically on the overhead tank and through a small hole the water level probe is inserted in the tank. A 0.5HP water pump is connected to the mechanical assembly, which will be controlled through the circuit. Working and testing of the circuit is shown in Fig. 11 through Fig. 12.

Case 1: Water level is below 75% and municipal supply is not available

- Water is filled through the pump and the red LED is on.
- Water filling stops when 75% water level has reached and now the yellow LED switches on.

Case 2: Water level is below 100% and municipal supply is available

- Water is filled through the municipal supply and the yellow LED is on.
- Water filling stops (solenoid valve turns off) when the tank becomes full and the green LED switches on.

Improvements possible

The project is by no means perfect and can be improved as per requirements. For instance, the water level probe can be further enhanced by adding a grid of metal contacts to precisely measure the water level.

Water remaining in the municipal supply lines, even when the supply is not available, sometimes gives false data about the availability of the supply. It can be improved by enhancing the mechanical design or by using a pressure sensor instead of metal electrode. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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